

To better understand the underlying mechanisms, this section will briefly deconstruct the critical components of TDOA positioning systems.

The goal of a TDOA system is to use the difference of time-of-arrivals of transmitted signals from multiple receiver stations to estimate the position of one or more transmitters. Receivers have fixed locations, while transmitters are assumed to be free to move.

For a signal $s(t)$ from a transmitter at time t_x , received replicas $r(t)$ can be assumed to be time shifted, scaled, and affected by AWGN (additive white gaussian noise) $n(t)$:

$$r(t) = As(t - \tau) + n(t).^1$$

Notice that for the difference in time of arrival $t_a - t_b$ between receiver station a and b, the transmission timestamp t_x cancels to zero, leaving only τ_a and τ_b as unknowns:

$$t_a - t_b = t_x + \tau_a - (t_x + \tau_b) = \tau_a - \tau_b.$$

Thus, the main objective from the receiver's standpoint is reduced to determining τ from a positioning signal from the transmitter. Existing literature varies in exact architecture to determine τ from $r(t)$, with differences arising from equipment cost, processing power, and power consumptions.² However, most come down to relying on engineering a proper $s(t)$ for estimating τ .

In an ideal scenario, a pulse function $r(t) = \delta(t - \tau)$ is received. Then,

$$\hat{\tau} = \underset{t \in \mathbb{R}}{\operatorname{argmax}} r(t).$$

Physically, this is impractical due to the infinite energy and bandwidth it would require. Though, it provides motivation for a sharp peak that can be resolved from a signal.

Furthermore, the achievable precision of this estimate is bounded by the Cramer-Rao Lower Bound (CRLB).³ Given a transmitted signal has Fourier transform $S(f)$, bandwidth B such that $S(f) = 0$ for $|f| > \frac{B}{2}$, zero-mean AWGN with variance σ^2 , and sampled at $f_s = B$, the variance of the unbiased estimator for τ is at least

$$\operatorname{Var}(\tau) \geq \frac{\sigma^2}{8\pi^2 f_s \int_{-f_s/2}^{f_s/2} f^2 |S(f)|^2 df}.^4$$

¹ S. Kruger, en“An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware,”.

² S. Kruger, en“An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware,”.

³ A. Leshchenko and S. Toledo, en“Modulation and Signal-Processing Tradeoffs for Reverse-GPS Wildlife Localization Systems,” in en2018 European Navigation Conference (ENC) (IEEE, Gothenburg, Sweden, 2018) pp. 154–165.

⁴ S. Kruger, en“An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware,”.

Notice that $AWGN \sim \mathcal{N}(0, \sigma^2)$ and $SNR = \frac{P_s}{\sigma^2}$, so, the CLRB is inversely proportional to both the SNR and signal bandwidth. The lower the CLRB, the better the delay estimate would be. Thus, a desirable signal should have high SNR and bandwidth.

To solve multiple of these design requirements, direct sequence spread spectrum (DSSS) can be used with a pseudorandom noise (PRN) code that has good autocorrelation properties to increase the signal's bandwidth, make it more resistant to interference, and enables code division multiple access (CDMA).⁵

Gold codes have good autocorrelation properties that include having a sharp peak, low sidelines, and low cross-correlation with other signals and members of the same family. Because of these useful properties, many designs from TDOA to global navigation satellite systems use Gold codes. Written out, for a PRN code $c[n] \in \{0, +1\}$ of length N , with a symbol, called a chip, at index n , the discrete autocorrelation at lag k should be, and have qualities of,

$$R[k] = \sum_n^{N-1} c[n]c[(n+k)mod(N)]$$

$$R[0] = N \text{ and } |R[k]| \ll N \forall k \neq 0.$$

Then, a suitable modulation method onto the carrier frequency is chosen between binary phase-shift keying (BPSK), OOK, and FSK. With these new definitions, the received baseband signal after carrier removal is expressed as

$$r(t) = Ac(t)e^{j(2\pi\Delta ft + \Delta\phi_0)}.$$

Where A is the amplitude, $c(t)$ is the continuous case of $c[n]$, with a total period of T_d and chips with *rect* functions of period T_c , Δf is the residual frequency mismatch mainly caused by clock differences⁶, and $\Delta\phi_0$ is the initial phase offset.

Next, a matched filter is implemented with the same Gold code as the transmitted signal. The correlator would output a peak S at lag $\tau = 0$ described by the expression

$$S = AT_d \text{sinc}(\Delta f T_d) e^{j(\Delta\phi_0 + \pi\Delta f T_p)}.$$

The magnitude of S is dependent on the frequency offset Δf . Low-cost commercial off the shelf (COTS) components usually cannot achieve required clock accuracy. Consequently, search algorithms can be implemented for received carrier frequency, and using this the received signal or locally stored template can be frequency shifted to minimize Δf .⁷

The index p at the correlator's peak is

⁵ R. MacCurdy, R. Gabrielson, E. Spaulding, A. Purgue, K. Cortopassi, and K. Fristrup, en"Real-Time, Automatic Animal Tracking Using Direct Sequence Spread Spectrum," .

⁶ S. Kruger, en"An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware," .

⁷ A. Bensky, enWireless positioning technologies and applications, GNSS technology and applications series (Artech House, Boston, Mass, 2008).

$$p = \underset{k \in N}{\operatorname{argmax}} S[k].$$

Since correlator output is in discrete samples, the true value of the peak (sub-index) is calculated based off Gaussian or parabolic peak interpolation. In different systems, this sub-index is defined as either an offset $\delta_p = \{-0.5, +0.5\}$ or timestamp that accounts for the offset. Using the differences between sub-indexes, which reflect the time of arrival, the position of the tags can then be calculated.

The preceding analysis treats the signal as given and asks how the receiver can estimate τ . The remainder of the section turns discussing different modulation schemes.

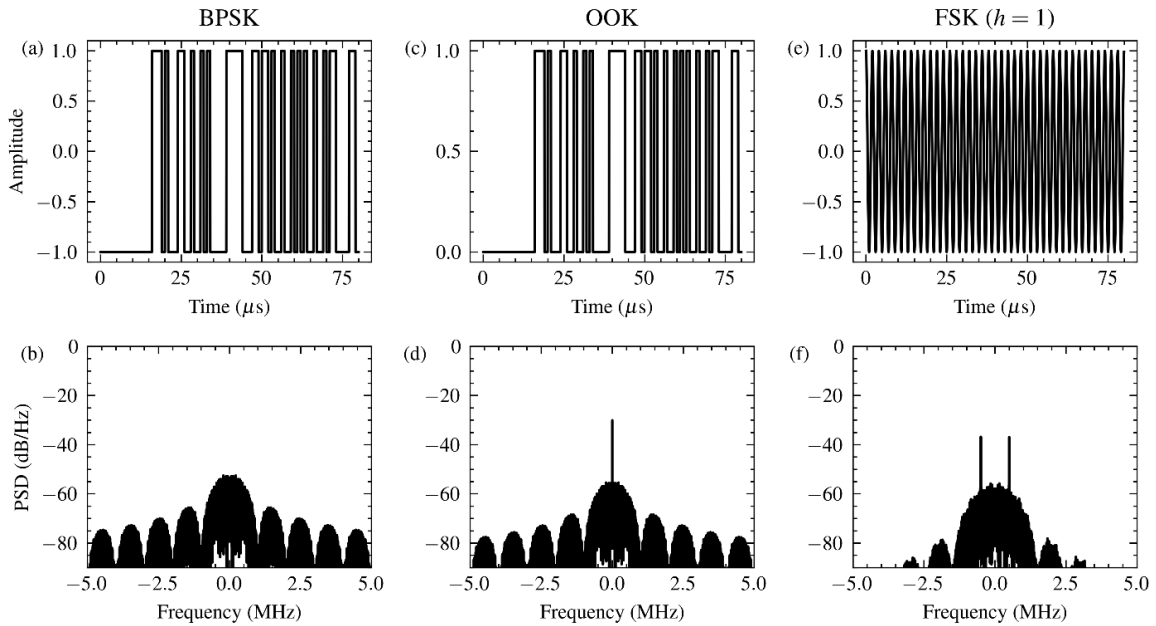


FIG. 1. Baseband waveforms (top row) and normalized power spectral densities (bottom row) for a 2047-chip Gold code at a 1 MHz chip rate under three modulation schemes. (a), (b) BPSK; (c), (d) OOK; (e), (f) discontinuous-phase FSK ($h = 1$), modeled as two free-running oscillators at ± 500 kHz gated by the code. Time-domain panels show the first 80 chips; the FSK waveform is the real part of the complex baseband signal. The PSDs are normalized to unit total power: BPSK shows a squared-sinc envelope with no discrete carrier line, OOK has a carrier line at 0 Hz containing about half the total power, and FSK shows two squared-sinc lobes at ± 500 kHz separated by 1 MHz.

BPSK represents binary symbols with a 180° carrier phase shift. For a Gold code modulated with BPSK, its carrier is sufficiently suppressed, as seen in Fig. 1 (b). This makes BPSK attractive for TDOA because coherent receivers can perform complex matched filters on a known code, producing sharp timing peaks and high resilience against white noise and other-tag interference. For this reason, when carrier frequency is recovered, it is theoretically the best modulation scheme for TDOA systems.⁸ However, carrier frequency recovery is hard to carry out accurately in real-time due to its suppressed carrier.⁹

⁸ A. Leshchenko and S. Toledo, en“Modulation and Signal-Processing Tradeoffs for Reverse-GPS Wildlife Localization Systems,” in en2018 European Navigation Conference (ENC) (IEEE, Gothenburg, Sweden, 2018) pp. 154–165.

⁹ S. Kruger, en“An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware,” .

OOK represents symbols with the presence or absence of the carrier. In an OOK modulated random code, roughly half the signal's power is allocated to the carrier, seen in Fig. 1 (d), which makes carrier detection and frequency recovery easier. The allocated energy in the carrier does not contribute to signal detection, reducing the timing energy efficiency compared to BPSK. This also slightly reduces the SNR, and thus accuracy of the estimates, in exchange for faster processing speeds. Despite these downfalls, $\pm 6m$ accuracy has been achieved with this modulation scheme.¹⁰

FSK represents symbols using two carrier frequencies f_- and f_+ separated by Δf around carrier frequency f_c . Since FSK occupies more bandwidth than OOK or BPSK at the same symbol rate, its larger frequency separation could reduce CRLB in an ideal coherent receiver. However, this advantage is not guaranteed. If the FSK waveform is discontinuous in phase, the receiver cannot fully exploit carrier phase in a complex matched filter, resulting in a reduction in correlation peak magnitude, and thus TDOA accuracy, despite the occupied bandwidth increase. Thus, discontinuous FSK can be spectrally inefficient¹¹ while still producing worse practical TDOA accuracy than coherent BPSK. Nevertheless, FSK is common among wildlife tags and supported by many RF chips. ATLAS series demonstrate that this trade off can still be field-effective, achieving meter-scale accuracy with kilometer scale receiver networks.¹²

For all modulation schemes seen in Fig. 1, the resulting PSD shape of the modulated Gold code is described by $T_c \text{sinc}^2(T_c f)$, where $\frac{1}{T_c}$ is the chip rate. At a chip rate of 1 MHz, the main lobe spans 2 MHz in bandwidth. Having higher chip rate increases the bandwidth of the resulting signal, which makes it more resilient to noise, and allows for more accurate sub-indexes from peak interpolation.

¹⁰ S. Kruger, en“An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware,” .

¹¹ S. Kruger, en“An inexpensive hyperbolic positioning system for tracking wildlife using off-the-shelf hardware,” .

¹² O. Kishon and S. Toledo, en“The ATLAS Wildlife LocalizationSystem: System Design and Implementation” .